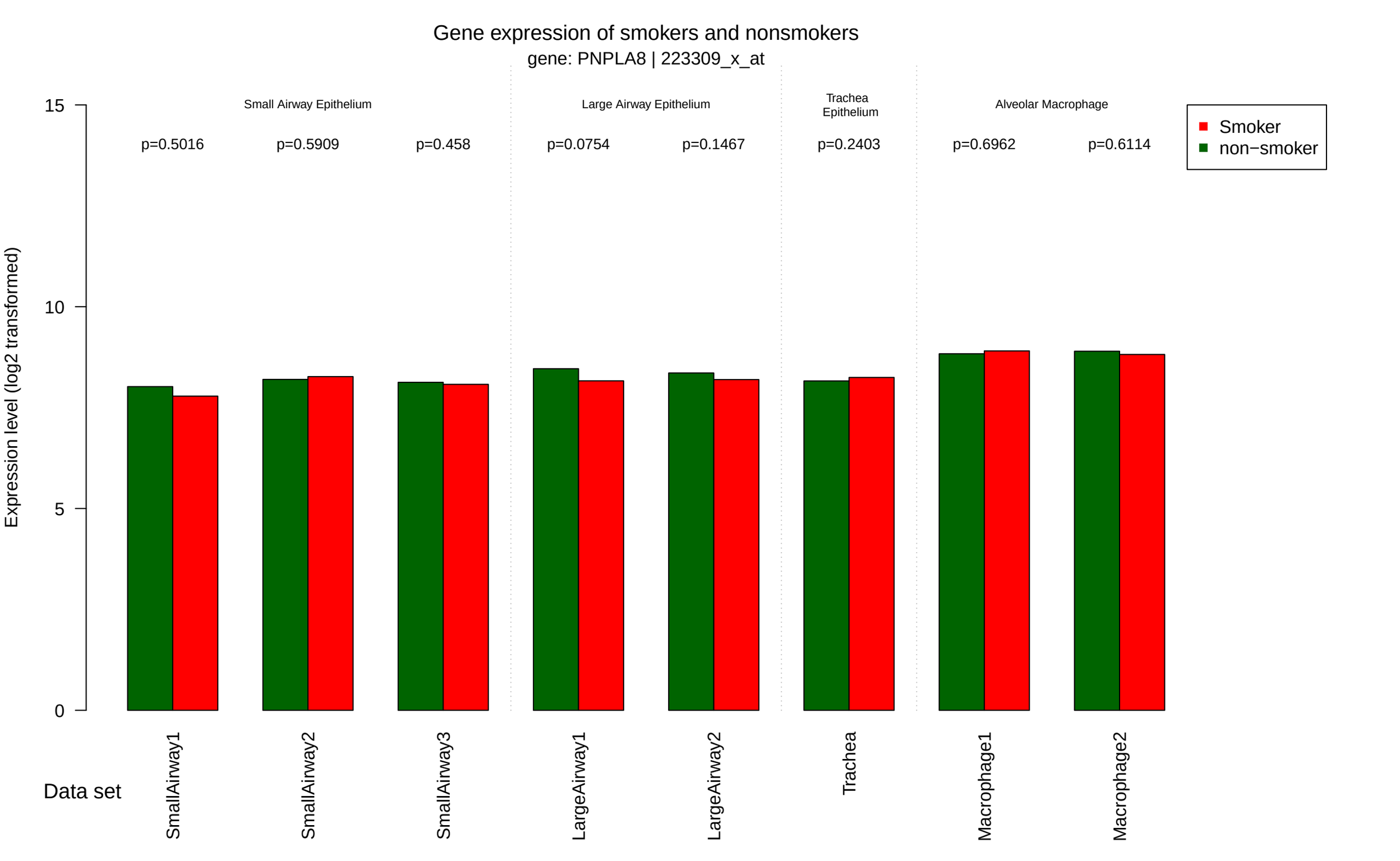
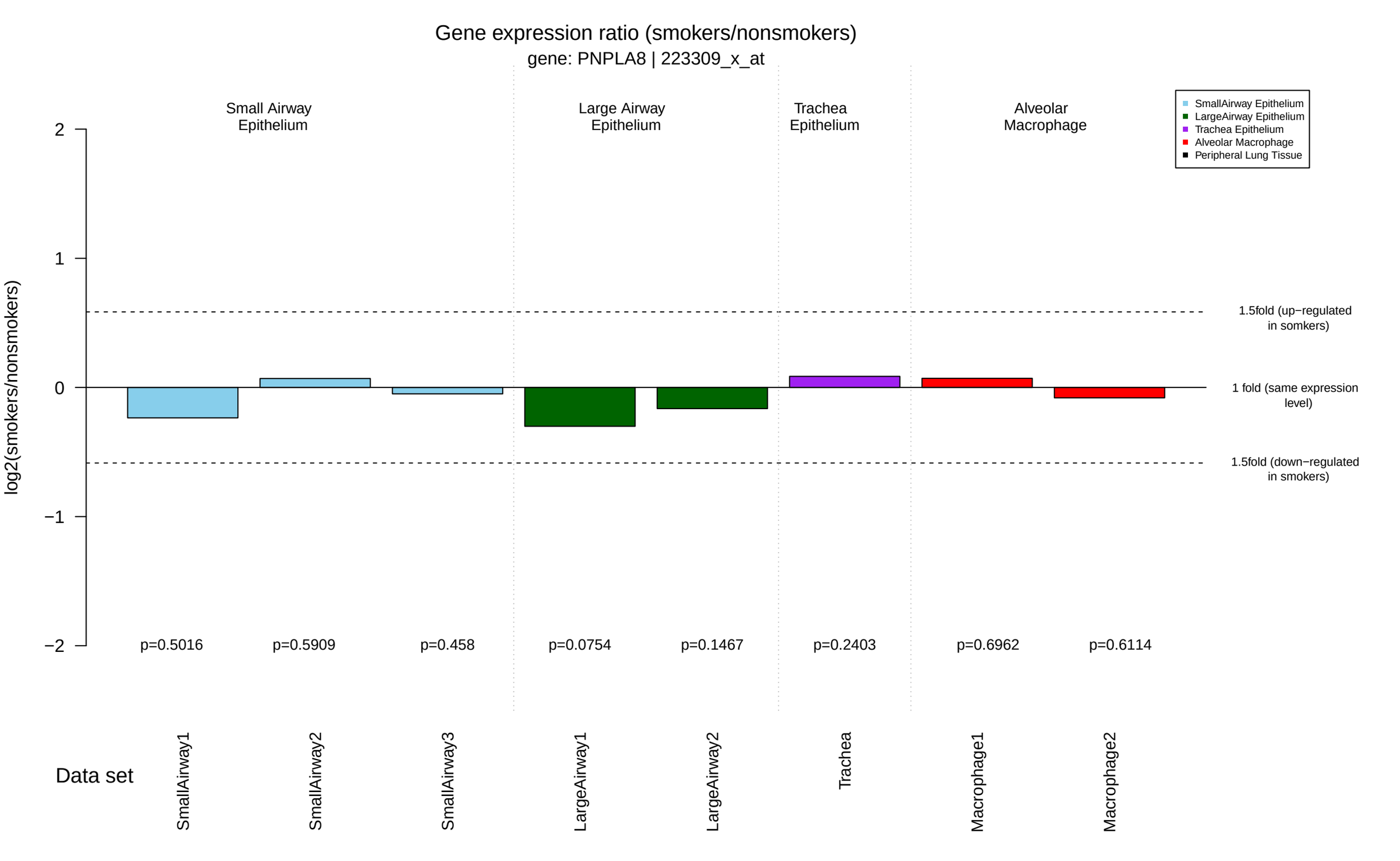
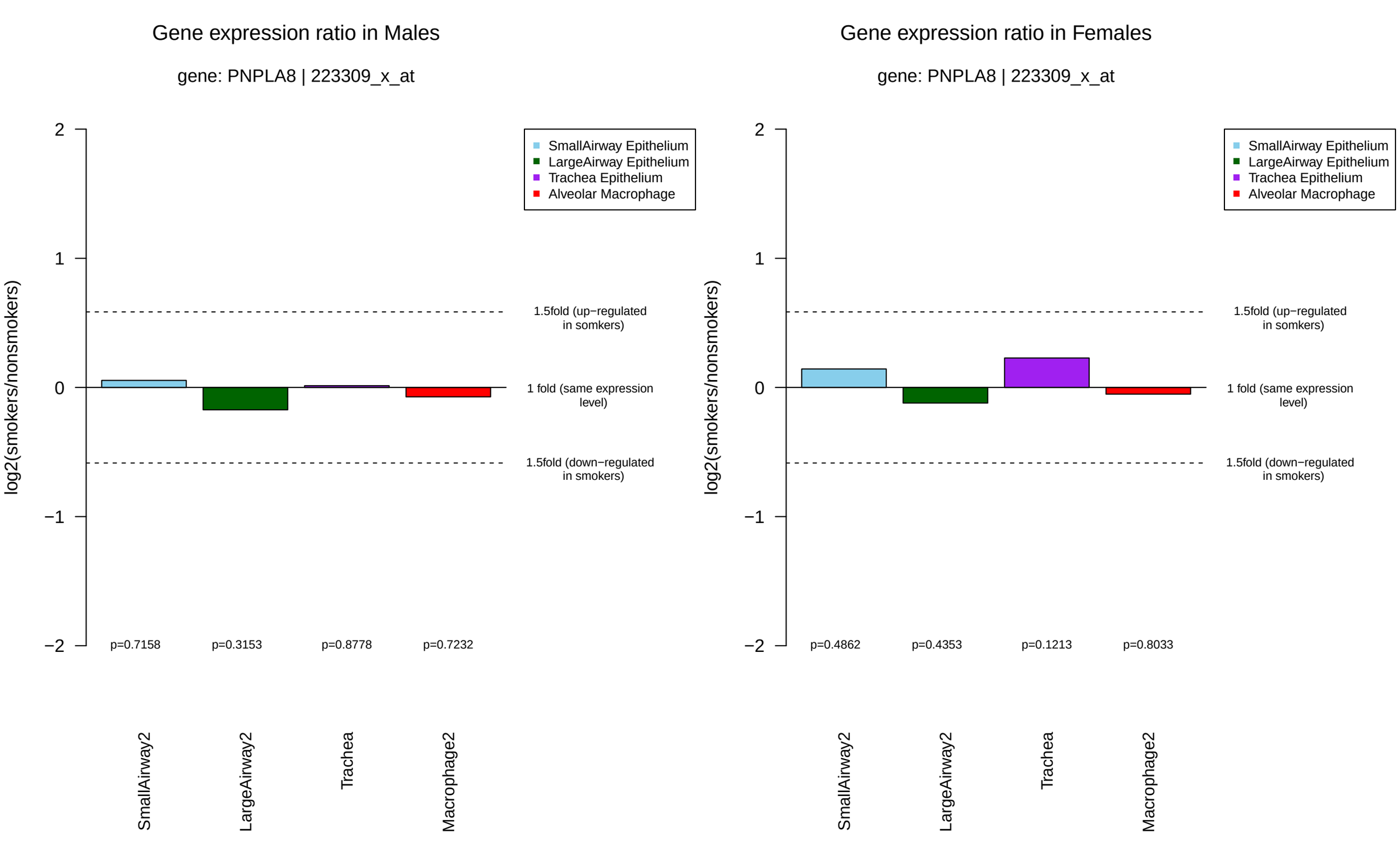
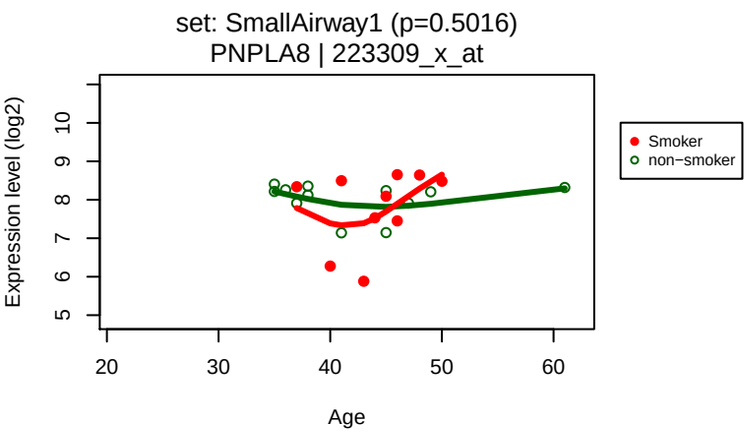
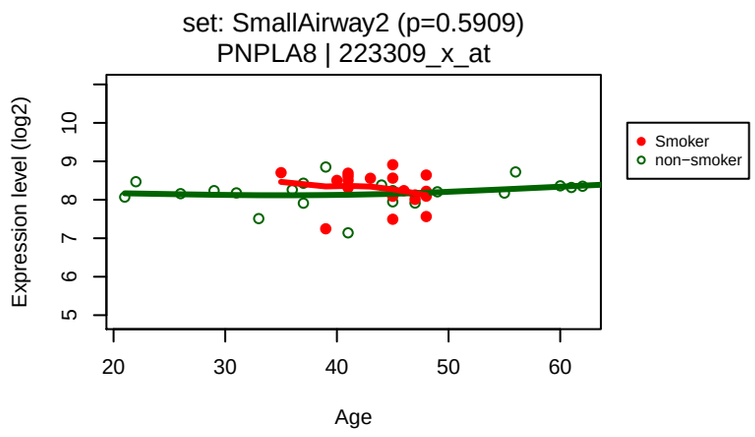
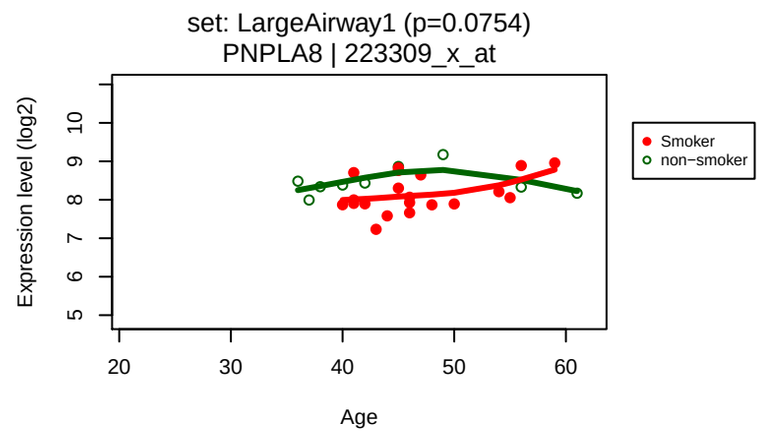

Gene expression of smokers and nonsmokers
gene: PNPLA8 | 223309_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5016
p=0.5909
p=0.458
p=0.0754
p=0.1467
p=0.2403
p=0.6962
p=0.6114
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

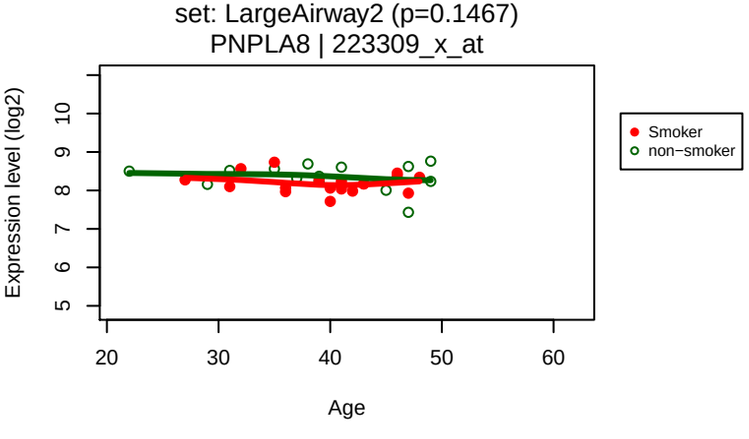

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA8 | 223309_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5016
p=0.5909
p=0.458
p=0.0754
p=0.1467
p=0.2403
p=0.6962
p=0.6114
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

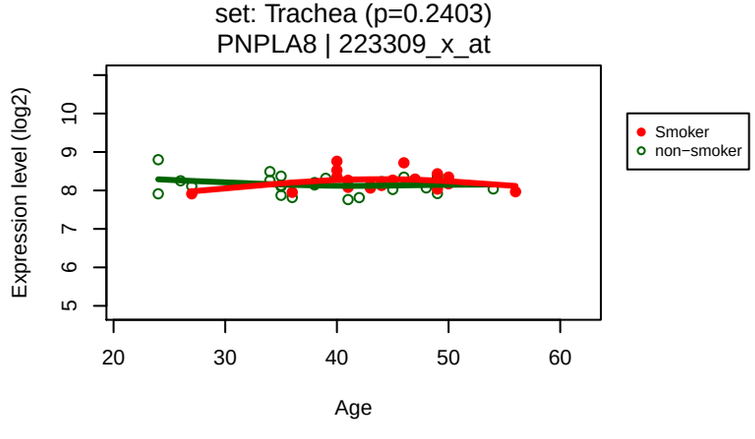

Gene expression ratio in Males
gene: PNPLA8 | 223309_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7158
p=0.3153
p=0.8778
p=0.7232
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA8 | 223309_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4862
p=0.4353
p=0.1213
p=0.8033
SmallAirway2
LargeAirway2
Trachea
Macrophage2

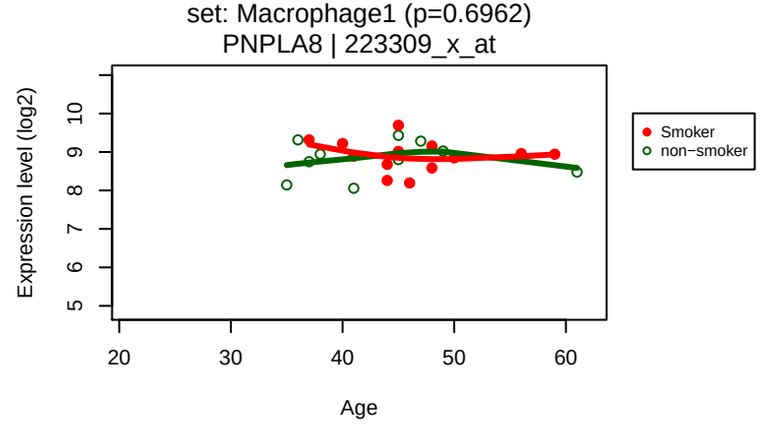

set: SmallAirway1 (p=0.5016)
PNPLA8 | 223309_x_at
Expression level (log2)
Age
Smoker
non−smoker

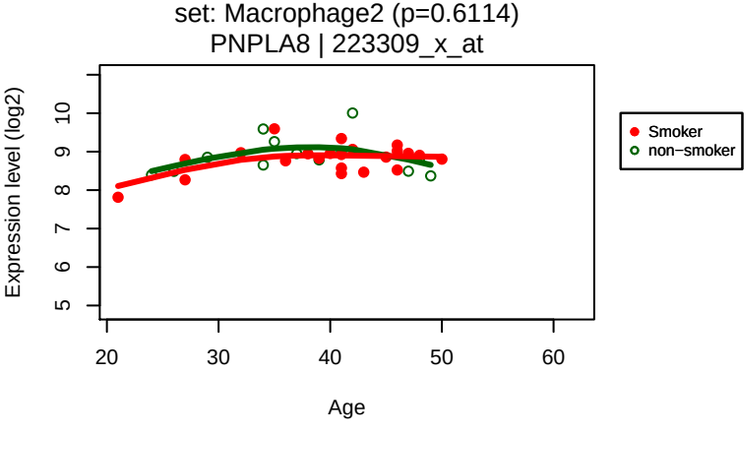

set: SmallAirway2 (p=0.5909)
PNPLA8 | 223309_x_at
Expression level (log2)
Age
Smoker
non−smoker

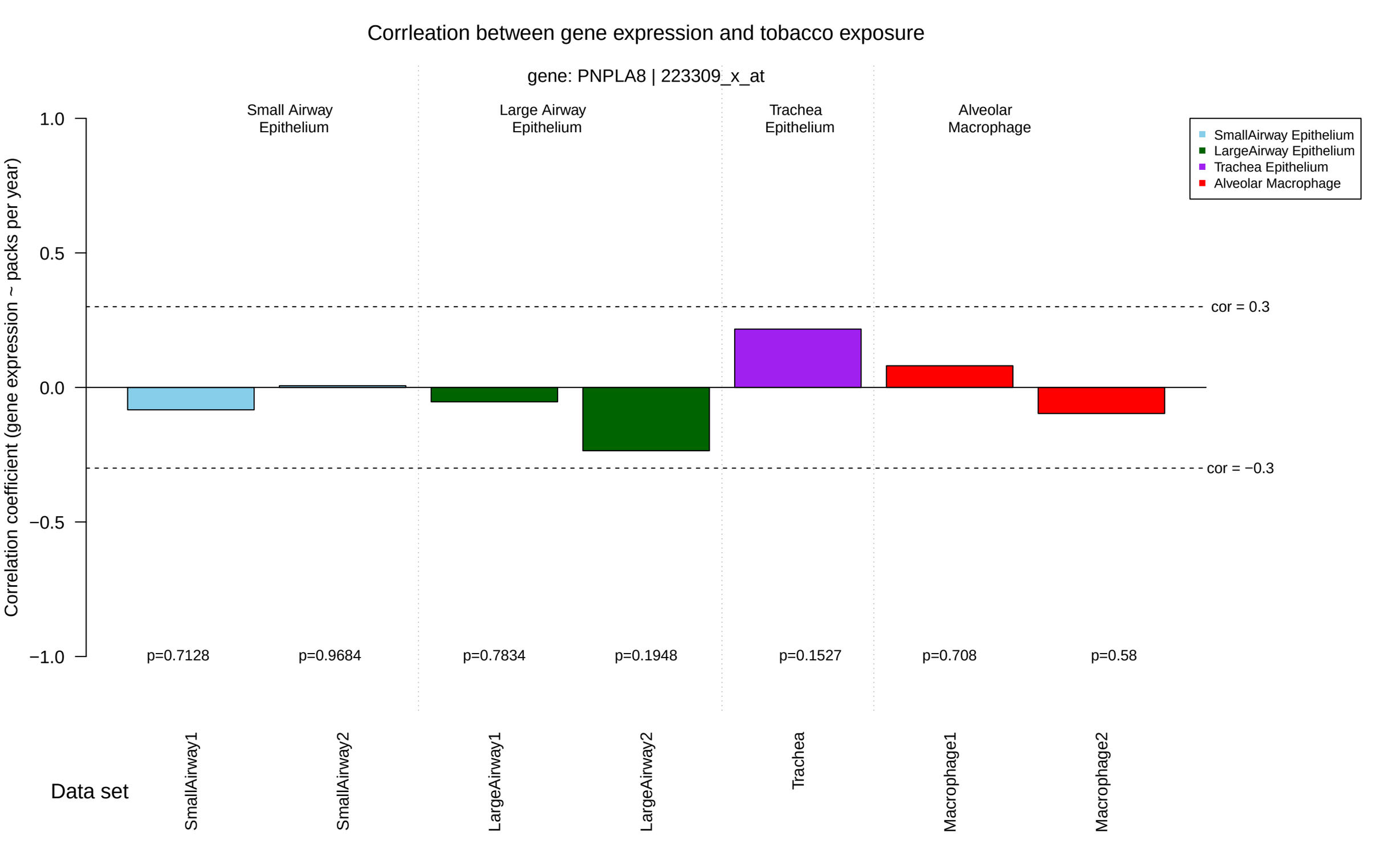

set: LargeAirway1 (p=0.0754)
PNPLA8 | 223309_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1467)
PNPLA8 | 223309_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2403)
PNPLA8 | 223309_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6962)
PNPLA8 | 223309_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6114)
PNPLA8 | 223309_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA8 | 223309_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7128
p=0.9684
p=0.7834
p=0.1948
p=0.1527
p=0.708
p=0.58
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2